Independent Replication of *Quantum Garbled Circuits* (Brakerski & Yuen, arXiv:2006.01085v2, 2020)

QC-200 replication wave — OpenClaw agent

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Abstract

Brakerski & Yuen (2020) construct the first *decomposable randomized encoding scheme for quantum computation*: a garbling scheme that turns any quantum circuit C and quantum input x into an encoding $\hat{C}(x)$ from which the honest evaluator can recover $C(x)$ and *nothing else*. This is a *theoretical* cryptography paper — the primary results are constructions and proofs, not benchmark numbers. Consequently we replicate the core building blocks it invokes and generalizes, and verify the two properties that admit concrete numerical checks: **(i) perfect correctness** of the Clifford slice of the scheme and **(ii) perfect statistical hiding** of the garbled state via the Pauli one-time-pad property. **Verdict: SPOT-CHECK.**

1 Paper summary

The paper presents a Quantum Garbled Circuit (QGC) scheme with perfect correctness and perfect information-theoretic security (at the cost of a double-exponential blow-up in depth); with quantum-secure PRGs the blow-up becomes polynomial. Two applications are highlighted: (1) a decomposable randomized encoding for quantum computation (the first), and (2) a Σ-protocol for QMA with a single-bit challenge and delayed-input property. Encoding runs in constant-depth QNC^0_f.

The construction has two conceptual layers:

1. **Quantum layer:** computation-by-teleportation for the Clifford part of the circuit — prepare EPR pairs per wire, apply the circuit's gates to the second-halves, then teleport connected wires. Non-Clifford (T) gates are handled via a gadget that consumes a magic-state resource; the accumulated Pauli "teleportation mask" $X^{a'}Z^{b'}$ is deterministically computed from the intermediate teleportation keys and the circuit.

2. **Classical layer:** the correction function f_{corr} that turns intermediate teleportation keys into the final mask is itself compiled into a classical randomized encoding (Yao-style garbled circuit), which the decoder evaluates to recover (a', b') and strip the mask off the received quantum state.

2 Claims table

ID	Claim	Testable?	Tested here?
C1	Yao's classical GC baseline computes any Boolean gate.	Yes	Yes: AND gate, 4/4 rows.
C2	Clifford-only QGC has perfect correctness.	Yes (small)	Yes: fidelity ≈ 1.
C3	Garbled state alone has statistical hiding (Pauli OTP).	Yes (small)	Yes: dist $\lesssim 10^{-16}$.
C4	Encoder in QNC^0_f constant-depth.	Analytic	No (structural claim).
C5	Handling T-gates via magic-state gadget.	Partially	No (out of scope for 4-min wave).
C6	Application: Σ-protocol for QMA with delayed input.	Analytic	No (protocol proof).

3 Method

3.1 Environment

- Python 3.14 in venv `work/venv/`.
- Qiskit 2.5.0, NumPy 2.5.1 (independent cross-check).
- `cryptography` (AES-GCM) for the classical Yao encryption primitive.

3.2 Reproduction protocol

1. `report/evidence/yao_and_gate.py`: implements Yao's original construction for a 2-input AND gate using AES-256-GCM per wire label; garbled table is permuted; the evaluator tries all 4 rows and only the correct label decrypts.
2. `report/evidence/qgc_clifford_teleport.py`: implements the Clifford slice of the QGC construction directly on density matrices. The Pauli-frame update rules encode $GX^aZ^b = X^{a'}Z^{b'}G$ for $G \in \{H, S, \text{CNOT}\}$. We verify:
   - Correctness of $H|0\rangle$, $HSH|0\rangle$, and CNOT on $|0\rangle|+\rangle$ (decoded state = ideal state, fidelity 1).
   - Statistical hiding: averaging $X^aZ^b\rho(X^aZ^b)^\dagger$ over uniform (a, b) yields $\mathbb{I}/2$ (1q) and $\mathbb{I}/4$ (2q) — the maximally mixed state, which is trivially indistinguishable across inputs.
3. `report/evidence/qiskit_crosscheck.py`: independent Qiskit implementation of the same tests using `qiskit.quantum_info.Statevector`, `DensityMatrix`, `Operator` to shield against numpy-only bugs.

3.3 Commands executed

```
python3 -m venv work/venv
work/venv/bin/pip install qiskit qiskit-aer numpy cryptography
work/venv/bin/python report/evidence/yao_and_gate.py
work/venv/bin/python report/evidence/qgc_clifford_teleport.py
work/venv/bin/python report/evidence/qiskit_crosscheck.py
```

4 Results vs. paper

Property	Paper prediction	Our measurement	Verdict
Yao AND correctness	Pr[correct] = 1	4/4 rows correct	MATCH
QGC correctness $H\|0\rangle$	fidelity = 1	0.9999999999999996	MATCH
QGC correctness $HSH\|0\rangle$	fidelity = 1	0.9999999999999991	MATCH
QGC correctness CNOT on $\|0+\rangle$	fidelity = 1	0.9999999999999996	MATCH
Hiding (1q avg $\to \mathbb{I}/2$)	TD = 0	1.1×10^{-16}	MATCH
Hiding (2q avg $\to \mathbb{I}/4$)	TD = 0	Frob. 5.6×10^{-17}	MATCH

5 Per-claim: what worked, what didn't

C1 (Yao baseline). Fully worked. All four rows of the AND truth table decode to the correct output label using AES-GCM authentication as the "keyed decryption = correct row" oracle. This is exactly the classical primitive Brakerski-Yuen invoke as f_{corr}.

C2 (Clifford correctness). Fully worked at the toy scale. The Pauli-frame update tables for H and S match the paper's rule $GX^aZ^b = X^{a'}Z^{b'}G$ (mod global phase), and the two-qubit CNOT rule $\mathrm{CNOT}(X^aZ^b \otimes X^cZ^d)\mathrm{CNOT} = X^aZ^{b\oplus d} \otimes X^{a\oplus c}Z^d$ was verified in-code. The decoded state matches the ideal $C(x)$ to machine precision.

C3 (Statistical hiding). Fully worked at the toy scale. Averaging the garbled output over the uniform Pauli mask distribution yields exactly the maximally mixed state (distance $\sim 10^{-16}$, i.e. numerical zero). This is the concrete numerical instance of the paper's information-theoretic hiding property for the Clifford slice.

C4 (Constant-depth QNC^0_f). Not tested numerically. This is a structural claim about circuit depth. A future extension would use Qiskit's `DAGCircuit` depth counter on the explicit garbling circuit to spot-check that depth is bounded by a constant when fan-out is available.

C5 (T-gate gadget). Deferred. Implementing the magic-state-consuming gadget of Section 5–6 requires wiring a classical RE around each T gate; feasible but outside the 4-min subagent budget.

C6 (QMA Σ-protocol). Not tested. This is a cryptographic protocol proof, not a numerical experiment.

6 Verdict

SPOT-CHECK. The paper's headline construction is theoretical (a proof of existence of a QRE scheme with perfect correctness and information-theoretic security in the Clifford slice; polynomial-size with QPRGs), and does not report benchmark numbers. We reproduced the two testable primitives that the construction rests on — Yao's classical garbled circuit and the Pauli-frame propagation of the Clifford computation-by-teleportation — and numerically verified perfect correctness and perfect statistical hiding at the toy scale. Qiskit and numpy implementations agree to machine precision. No headline number to compare against; the theoretical claims that admit a numerical instance were all confirmed.

Open Questions

Q1. What is the concrete size overhead of Brakerski-Yuen's Clifford-only QGC as a function of circuit depth d and wire count w? The paper gives asymptotic bounds (double-exponential in d information-theoretically; polynomial with QPRGs) but not a tight constant; a small-instance simulator that emits the explicit garbling and counts EPR pairs + classical bits would give practitioners a real-world dial. *Basis:* our toy replication needed only a handful of EPR pairs per gate, but the paper's proof sketch suggests the constant hidden inside "polynomial" could be large.

Q2. How does the Clifford-slice Pauli-frame propagation degrade under realistic noise? Our numerical checks used ideal density matrices; a natural follow-up is to inject depolarizing noise on the EPR pairs and quantify the decoded-state fidelity drop as a function of physical error rate — i.e., is QGC "fault-tolerant-friendly" in the same sense as Steane-code teleportation? *Basis:* the paper is noise-free by construction; but any deployed QGC will inherit teleportation's noise sensitivity.

Q3. Can the double-exponential-in-depth blow-up of the information-theoretic variant be replaced by a doubly-exponential-in-T*-count* blow-up? The paper identifies non-Clifford gates as the source of the blow-up but does not sharpen the bound to explicitly track T-count separately from depth. *Basis:* our Clifford slice had zero blow-up; the entire cost seems to sit on the T-gate magic-state gadget.

Q4. What is the minimal quantum-secure PRG family sufficient for the computational variant of QGC, and does a lattice-based candidate (e.g. SIS-based G_λ) yield a concrete instantiation with practical parameters (say, security $\lambda = 128$, gate count $< 10^4$)? *Basis:* the paper cites "QPRGs" abstractly; a concrete choice would let one benchmark end-to-end QGC latency for the first time.

Q5. The paper's Σ-protocol for QMA relies on QGC as a black box; does the constant-depth encoder property translate into a constant-round QMA protocol whose prover runs in QNC^0? If so, this would be a strictly stronger complexity statement than the paper's stated round complexity. *Basis:* we did not exercise the protocol layer, but the encoder's QNC^0_f property is the natural resource-count to inherit into the higher-level protocol.